TradingResearch Pro

Research & Analysis Reference

Last updated: 2026-06-20 | Version 2.0

Covers: research.py · backend/services/stock_analyzer.py · sentiment.py · backend/services/regime_detector.py

1. Overview — Two Analysis Paths

The app has two distinct analysis paths that share data sources but serve different purposes:

Feature	File	Mode	Purpose
Research	research.py	Free (yfinance) or API (Claude Opus)	Rank a universe of assets by opportunity score; generate daily picks report
Stock Analysis	backend/services/stock_analyzer.py	Always uses Claude Sonnet	Deep single-stock analysis with full technical + fundamental AI narrative

Since the recent update, Research computes and displays the same named technical indicators (RSI, MACD, Bollinger, SMA, VWAP, ATR) that Stock Analysis shows. Technical indicator values are also injected into the Claude deep-research prompt via *_build_technical_context()* so the AI has quantitative data to reason from.

2. Market Regime Detection

File: backend/services/regime_detector.py | **Sources:** VIX (^VIX), SPY — fetched daily, cached 1 hour | **Used by:** Both Research scoring and Stock Analysis AI prompt

The regime classifier sets a **score multiplier** applied to every asset's composite score after all sub-scores are combined.

Regime	Condition	Multiplier	Effect
BULL	VIX ≤ 18 AND SPY > SMA50 by >1% AND > SMA200 by >5%	1.10×	Scores boosted 10%
NEUTRAL	Default (mixed signals)	1.00×	No adjustment
BEAR	VIX ≥ 25 OR (SPY < SMA50 by 3% AND < SMA200 by 5%)	0.75×	Scores dampened 25%
CRISIS	VIX ≥ 35 OR (VIX ≥ 28 AND deep drawdown)	0.60×	Scores dampened 40%

The regime is also passed to the Stock Analysis AI prompt to contextualise position sizing advice (*favour tight stops in a BEAR/CRISIS environment*).

3. Technical Indicators

3a. Research (Free Mode) — research.py

Computed inside *_score_asset()* from 3-month daily OHLCV history (yfinance). All values appear in the email HTML under each pick and are injected into Claude prompts.

Indicator	Parameters	Display / Notes
RSI	Period: 14	< 30 = oversold (green), > 70 = overbought (red)
MACD	Fast 12, Slow 26, Signal 9	Line, signal line, histogram shown
SMA 20	20-day simple moving average	Short-term trend
SMA 50	50-day simple moving average	■ price above, ■ price below
SMA 200	200-day simple moving average	null if < 200 days of history available
Bollinger Bands	Period 20, Std dev 2×	Upper / lower / mid band
VWAP	Full 3-month window	Volume-Weighted Average Price; ■ if price above
ATR	Period: 14	Average True Range shown as % of price (volatility)

3b. Stock Analysis — backend/services/stock_analyzer.py

Computed from user-selected time period (1d / 1w / 1m / 3m / 6m / 1y). RSI period, Bollinger period/std, and MACD parameters are all user-configurable from the UI.

Indicator	Default Params	User-configurable
RSI	Period: 14	Yes — RSI period
MACD	Fast 12, Slow 26, Signal 9	Yes — all three periods
SMA 20	20-day	No (fixed)
SMA 50	50-day	No (fixed)
SMA 200	200-day	No (fixed)
Bollinger Bands	Period 20, Std 2.0	Yes — period and std multiplier
VWAP	Full selected period	No (always computed)

Indicator	Default Params	User-configurable
ATR	Period: 14	No (fixed)

Technical Signal Score (Stock Analysis only)

Derived from indicator values to produce a signal (BUY / WATCH / HOLD / SELL) and a 0–100 score. Confidence = % of active indicator votes aligned with the final signal.

Indicator Condition	Bullish?	Score Impact
RSI < 30 (oversold)	Yes	+15
RSI < 45	Yes	+7
RSI > 70 (overbought)	No	–15
RSI > 55	No	–5
MACD > Signal line	Yes	+10
MACD < Signal line	No	–10
Price > SMA 50	Yes	+8
Price < SMA 50	No	–8
Price > SMA 200	Yes	+7
Price < SMA 200	No	–7
Day change > 0	Proportional	±up to 10
Volume ratio > 1.5×	Yes	+5
Price < Bollinger Lower	Oversold bounce	Vote only
Price > Bollinger Upper	Overbought reversal	Vote only
Price > VWAP	Yes	+6
Price < VWAP	No	–4

Final signal: BUY (score ≥ 65) / WATCH (52–64) / HOLD (38–51) / SELL (< 38)

4. Fundamental Factors

4a. Research Scoring Fundamentals

Fetched from yfinance Ticker.info for each asset.

Factor	yfinance Field	How Used in Scoring
Forward EPS growth	forwardEps vs trailingEps	EPS growth sub-score
Earnings Growth	earningsGrowth	Fallback if forward EPS unavailable
Revenue Growth	revenueGrowth	Part of earnings quality score (20% weight)
EPS Surprise	earnings_history (surprise%)	Beat (+) / miss (–) vs analyst estimate
Short % of Float	shortPercentOfFloat	Squeeze potential or bearish signal
Short Ratio	shortRatio	Days to cover — used in short score
Sector	sector	Maps to sector ETF for relative strength

Earnings quality sub-score (0–100): 50% × EPS beat/miss + 30% × EPS growth + 20% × revenue growth

Short interest score logic:

- Short > 15% AND price rising → squeeze setup (score 70–92)
- Short > 30% AND price falling → confirmed short thesis (score 22)
- Short > 20%, no direction → headwind (score 40)
- Default neutral → score 50

4b. Stock Analysis Fundamentals

Field	Description
Company name, sector, industry	Identity
Market cap	Size classification
P/E ratio (trailing & forward)	Valuation
EPS (trailing & forward)	Earnings power
EPS growth (fwd vs trailing)	Growth trajectory
EPS surprise (last quarter)	Beat / miss vs consensus estimate
Revenue & revenue growth (YoY)	Top-line momentum
Profit margin	Profitability
Debt-to-equity	Balance sheet risk
Current ratio	Liquidity
Return on equity (ROE)	Capital efficiency
Dividend yield	Income
Beta	Market correlation / volatility
Short % of float + short ratio	Short interest

5. Sentiment Factors

File: sentiment.py | **Cache:** 30 minutes in-process | **Weight in composite:** 5% | **Used by:** Research scoring + API prompt injection

Source	Type	API Key?	What It Provides
StockTwits	Per-ticker	No	Bullish/bearish message counts (last 30 messages)
CNN Fear & Greed	Macro (stocks)	No	Score 0–100; cached per session
Alternative.me	Macro (crypto)	No	Crypto Fear & Greed score 0–100; crypto assets only
Reddit	Per-ticker	Optional	Post count + bull/bear keyword ratio across WSB, stocks, investing, StockMarket, pennystocks, cryptocurrency

Reddit requires REDDIT_CLIENT_ID and REDDIT_CLIENT_SECRET in .env to activate.

Sentiment Score Blending:

Data Available	Formula
StockTwits + Fear&Greed; + Reddit	50% StockTwits + 35% Fear&Greed; + 15% Reddit
StockTwits + Fear&Greed; (no Reddit)	60% StockTwits + 40% Fear&Greed;
Fear&Greed; only	100% Fear&Greed;

Output labels: Bullish (≥65) / Leaning Bullish (55–64) / Neutral (45–54) / Leaning Bearish (35–44) / Bearish (<35)

6. Insider & Institutional Data

Source: yfinance Ticker.insider_transactions (SEC Form 4) + Ticker.major_holders | **Lookback:** Last 90 days | **Weight in composite:** 5% | **Stocks only** (skipped for crypto)

Net Shares Bought (last 90 days)	Insider Score	Interpretation
> 100,000	85	Strong insider buying — very bullish signal
20,001 – 100,000	70	Moderate buying
1 – 20,000	58	Slight buying
0 (neutral)	50	No notable activity
–1 to –50,000	42	Slight selling
–50,001 to –200,000	33	Moderate selling
< –200,000	22	Heavy selling — red flag

Automatic sales (Rule 10b5-1 plans) are excluded from the sell count.

Institutional Ownership Bonus:

Institutional Ownership	Score Bonus
> 75%	+8 points
50%–75%	+4 points
< 50%	No adjustment

7. Analyst Consensus

Source: yfinance Ticker.info | **Fields:** targetMeanPrice, recommendationMean (1=Strong Buy → 5=Sell), numberOfAnalystOpinions | **Weight in composite:** 10% | **Minimum analysts:** 2

Analyst Score Formula:

```
price_score = clamp(50 + analyst_upside_pct × 1.5, 0, 100)
rec_score = clamp(100 − (recommendationMean − 1) × 22.5, 0, 100)
reliability = min(num_analysts / 8, 1.0)
analyst_score = (price_score × 0.5 + rec_score × 0.5) × reliability
+ 50 × (1 − reliability)
```

The reliability factor scales from 0→1 as analyst count goes from 0→8+, so a single analyst opinion does not dominate the score.

recommendationMean	Label	rec_score
1.0	Strong Buy	100
1.5	Buy	88.75
2.0	Buy	77.5
2.5	Hold	66.25
3.0	Hold	55
3.5	Underperform	33.75
4.0	Underperform	22.5
5.0	Sell	0

8. Earnings Calendar Awareness

Source: yfinance Ticker.calendar | **Lookback:** Next 7 days | **Applied as:** Score penalty after composite × regime multiplier

Stocks with imminent earnings are flagged and penalised because the outcome is binary and unpredictable — holding into earnings is speculation, not analysis.

Days to Earnings	Score Penalty	Display
0–1 day	−15 points	■ Earnings tomorrow
2–3 days	−10 points	■ Earnings in X days
4–7 days	−5 points	■ Earnings this week
> 7 days	0	(no flag)

Formula: score = composite × regime_multiplier − earnings_penalty

9. Composite Scoring — Research Free Mode

Function: _score_asset() in research.py | **Scale:** 0–100 | **Signal:** BUY ≥ 65 / WATCH 45–64 / HOLD < 45

Sub-score Formulas (each normalised 0–100):

Sub-score	Formula
mom_1d	day_change_pct × 5 + 50
mom_1w	week_change_pct × 3 + 50
mom_1m	month_change_pct × 2 + 50
mom_3m	qtr_change_pct × 1.5 + 50
vol_scr	min(vol_ratio × 40, 100)
pos_scr	52w_position × 100 (0 = at 52w low, 100 = at 52w high)
rs_spy	(qtr_change – SPY_3m) × 2 + 50
rs_sector	(qtr_change – sector_ETF_3m) × 2 + 50
earn_qual_scr	EPS_beat×0.5 + EPS_growth×0.3 + rev_growth×0.2
short_scr	See short interest logic in Section 4a
sent_scr	From sentiment.py (0–100)
analyst_score	See Section 7 formula
insider_score	See Section 6 table

Composite Weights (sum = 100%):

Factor	Weight
3-month momentum	18%
1-month momentum	14%
1-week momentum	9%
1-day change	2%
Volume surge	9%
Relative strength vs SPY	7%
Relative strength vs sector ETF	7%
52-week position	3%
Earnings quality (EPS + revenue)	7%
Short interest / squeeze potential	4%
Social sentiment	5%
Analyst consensus + price target	10%
Insider / institutional signal	5%
TOTAL	100%

Confidence score: percentage of sub-scores above 50 (how many factors agree the asset is above average):
confidence = count(sub_score > 50) / total_sub_scores × 100

10. AI Prompts

10a. Research — Simple Dual-Category Prompt (SYSTEM_PROMPT)

Used in run_api() for the All Stocks + Penny Stocks scheduled report mode. Model: claude-opus-4-8 with extended thinking and web_search tool.

```
You are a quantitative trading analyst. Research a given universe
of stocks and cryptocurrencies using live market data and news, then rank them by
short-term opportunity score for today.

IMPORTANT QUALITY BAR: Only include picks where you have 90%+ confidence.
Prefer returning fewer than 5 picks over including a low-conviction one.

For each asset, research ALL of the following:
1. Price momentum, volume vs 30-day average, and today's % change.
2. News sentiment from the last 48 hours.
3. Analyst consensus: current buy/hold/sell ratings and mean price target.
4. Earnings calendar: flag any earnings within the next 7 days (high event risk).
5. Insider activity: any notable Form 4 purchases or sales in the last 30 days.
6. Institutional ownership: is it rising or falling (13F signals)?
7. Technical signals and upcoming catalysts.

Penalise stocks reporting earnings within 3 days.
Treat recent insider buying as a strong positive; heavy selling as a red flag.

JSON output per pick: rank, ticker, asset_type, current_price, day_change_pct,
score, confidence_pct, signal, reasoning, key_catalyst, analyst_sentiment,
insider_activity, earnings_warning, suggested_entry, target_price, stop_loss,
time_horizon, risk_note

Signal: BUY | HOLD | WATCH. Score >= 90 to include. Rank descending.
```

10b. Research — Penny Stock Prompt (CHEAP_STOCK_SYSTEM_PROMPT)

Used in run_api() for penny stocks (price < $5).

```
You are a small-cap stock analyst specialising in low-price, high-potential
equities (under $5). Identify stocks that are cheap but have strong future
growth potential — NOT just momentum plays.

For each stock consider:
- Business model: real problem, viable path to profitability?
- Catalysts: earnings, product launches, FDA approvals, contracts
- Financial health: debt levels, cash runway, revenue trend
- News sentiment: positive developments in the last 7 days?
- Insider/institutional activity: any notable buying?
- Risk: temporary headwind or structural decline?

AVOID: pump plays, zero-revenue shells, terminal-decline companies.

JSON output per pick includes target_price, time_horizon (3-6 months),
why_its_cheap, business_viability, financial_health.
```

10c. Research — Sector Deep Research Prompt (DEEP_RESEARCH_PROMPT_TEMPLATE)

Used in run_sector_api() for sector-by-sector analysis. Sector guidance is injected per sector. Pre-computed technical indicators are also injected via _build_technical_context().

```
You are a senior equity research analyst at a top-tier investment fund.
Conduct rigorous, data-driven research on the {sector} assets provided.

SECTOR FOCUS: {sector_guidance}

RESEARCH PROCESS — execute ALL steps via web search:
1. Fetch live price, today's % change, volume vs 30-day average, 52-week range.
2. Search news from the last 7 days for each ticker.
3. Check analyst ratings and mean price target changes in the last 30 days.
4. Identify upcoming catalysts in next 60-90 days.
5. Review latest earnings: EPS beat/miss, revenue growth, guidance, margins.
6. Assess balance sheet: cash vs debt, free cash flow.
7. Check insider transactions (SEC Form 4) in the last 30 days.
8. Note institutional ownership trend — rising or falling.
9. Flag stocks with earnings within 7 days — reduce confidence accordingly.

QUALITY BAR: score >= 90 AND confidence >= 90%. Return UP TO {top_n} picks.

JSON output per pick: rank, ticker, company_name, current_price, day_change_pct,
week_change_pct, score, confidence_pct, signal, why_picked, technical_analysis,
fundamental_snapshot, key_catalyst, sector_tailwind, analyst_sentiment,
insider_activity, earnings_warning, news_summary, news_sentiment,
suggested_entry, target_price, stop_loss, upside_pct, time_horizon, risk_factors[]
```

Sector Guidance Injected Per Sector:

Sector	Key Focus Areas
Technology	AI/ML adoption, semiconductor demand, cloud ARR growth, software innovation, competitive moats, valuation vs growth
Pharma & Biotech	FDA calendar (next 90 days), Phase 2/3 trials, patent cliffs, pipeline value, M&A; activity, drug pricing
Healthcare	Insurance reimbursement, hospital utilization, medical device innovation, Medicare/Medicaid policy, managed care margins
Finance	Net interest margin, loan growth vs credit losses, capital adequacy, fee income, Fed rate trajectory
Energy	Crude/nat-gas futures curve, production guidance, refining spreads, capex discipline, OPEC+ decisions, energy transition
Consumer	Consumer confidence, same-store sales, gross margin recovery, inventory normalization, e-commerce, brand strength
Industrials	Order backlog, defense budget, infrastructure spending, supply chain, pricing power, international exposure
Crypto	On-chain metrics, institutional inflows, regulatory developments, network upgrades, DeFi activity

10d. Research — Penny Stocks Sector Prompt (PENNY_DEEP_RESEARCH_PROMPT_TEMPLATE)

```
You are a small-cap and penny stock specialist.
Research the provided stocks (all priced under ${max_price}) for
high-conviction recovery or growth plays.

KEY QUESTIONS FOR EACH STOCK:
- Why is it cheap? Temporary headwind or structural decline?
- Is the business viable? Real revenue, path to profitability?
- What catalyst could re-rate it?
- Financial runway: cash months, revenue trajectory, debt burden.
- Market interest: insider buying, institutional accumulation, short squeeze?

AVOID: pump plays, zero-revenue shells, terminal decline, fraudulent operators.

JSON output per pick includes: why_its_cheap, business_viability,
key_catalyst, financial_health, technical_analysis, target_price,
stop_loss, upside_pct, time_horizon, risk_factors[]
```

10e. Stock Analysis — AI Narrative Prompt

Used in _ai_analysis() in backend/services/stock_analyzer.py. Model: claude-sonnet-4-6 (max 600 tokens — concise and actionable).

```
You are a professional equity analyst. Analyze {ticker} ({company})
based on the following data and provide a concise, actionable analysis.

TECHNICAL DATA:
- Current Price / Day Change / RSI (14) / MACD
- 50-day SMA / 200-day SMA / VWAP (above/below — bullish/bearish intraday)
- ATR (14) as $ and % of price / Momentum Score: {score}/100
- Signal: {BUY | WATCH | HOLD | SELL}

FUNDAMENTAL DATA:
- Sector / P/E (trailing & forward) / Beta
- Profit Margin / Revenue Growth (YoY)
- EPS: trailing → forward ({growth}% expected)
- EPS Surprise (last quarter) vs estimate
- Short Interest: % of float + days to cover

ANALYST DATA:
- Consensus rating ({num_analysts} analysts)
- Price Target (mean): ${target} ({upside}% upside)
- Target Range: ${low} – ${high}

MARKET REGIME:
- Regime: {BULL|NEUTRAL|BEAR|CRISIS} (VIX={vix})
- SPY vs SMA50 / SMA200
- Score multiplier + regime-appropriate sizing advice

Provide:
1. Summary (2-3 sentences, includes regime context)
2. Technical Outlook (indicators + VWAP position)
3. Analyst Consensus (targets vs current price)
4. Key Risks (2-3 bullet points)
5. Trade Setup (entry, target, stop = 1.5× ATR, size for regime)
```

Appendix — Data Sources Summary

Data	Source	API Key?	Cache
Price, OHLCV, fundamentals	yfinance (Yahoo Finance)	No	Per request
Earnings calendar	yfinance Ticker.calendar	No	Per request
Analyst targets & consensus	yfinance Ticker.info	No	Per request
Insider transactions (Form 4)	yfinance insider_transactions	No	Per request
Institutional holders	yfinance major_holders	No	Per request
StockTwits sentiment	StockTwits public API	No	30 min
CNN Fear & Greed	CNN dataviz API	No	30 min
Crypto Fear & Greed	Alternative.me API	No	30 min
Reddit mentions	Reddit OAuth API	Optional	Per request
Market regime (VIX/SPY)	yfinance	No	1 hour
Deep research + web search	Claude Opus 4.8 + web_search	Required	Per run
Stock Analysis AI narrative	Claude Sonnet 4.6	Required	Per request

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